

Problem Set 2

Tate Mason

April 21, 2026

1 Introduction

This problem set was concerned with evaluating models of entry and competition. Given data from Jia (2008), I used a variety of estimation techniques to estimate the probability of entry. First, I used a probit, then a different specification of the probit, and finally an IV model to model entry simply. Next, I used an ordered probit á la Bresnahan and Reiss (1991) to model the effect of number of firms in a market on entry. Lastly, I used the Seim (2006) method to model the effect of predicted entry of a competitor on own entry.

This report is laid out as follows. Section 2 will present the profit and entry functions, section 3 will present the simple estimation results, section 4 will present the ordered probit results, and section 5 will present the Seim method results. Finally, section 6 will conclude.

2 Profit and Entry Functions

The profit function for Walmart is as follows:

$$\pi_t^W(N_t) = \underbrace{R_t(N_t)\bar{\mu}_t}_{\text{Net revenue}} - \underbrace{\left[\sum_i \sum_l D_{it}(N_t)\Omega_{il}(N_t)sh_{il}(N_t)\right]}_{\text{Shipping cost}} - \underbrace{\sum_l L(q_{lt})w_{lt}}_{\text{Labor cost}} - \underbrace{\sum_l F_{lt}}_{\text{Fixed cost}}$$

Where:

- $D_{it} = \frac{R_{it}(N_t)}{\bar{p}_t}$ is orders from county i
- $\bar{m}u_t$ is the average markup in the market
- $\Omega_{il}(N_t)$ is the order flow matrix
- $sh_{il}(N_t)$ is the cost of shipping from location l to county i
- $q_{lt} = \sum_i \Omega_{il}(N_t)D_{it}(N_t)$ is the orders fulfilled by location l

The profit function for KMart is as follows:

$$\pi_t^K(N_t) = \underbrace{R_t(N_t)\bar{\mu}_t}_{\text{Net revenue}} - \underbrace{\left[\sum_i \sum_l D_{it}(N_t)\Omega_{il}(N_t)sh_{il}(N_t)\right]}_{\text{Shipping cost}} - \underbrace{\sum_l L(q_{lt})w_{lt}}_{\text{Labor cost}} - \underbrace{\sum_l F_{lt}}_{\text{Fixed cost}}$$

Where all follows as above

The entry function for Walmart is as follows:

$$P(\text{Entry})^W = \max_a \sum_{t=0}^{\infty} (\rho_t \beta)^t \left[\sum_{j \in \mathcal{B}_t^K} (\pi_{jt}^g - c_{jt}^g - \tau d_{jt}^g) + \sum_{j \in \mathcal{B}^0} (\pi_{jt}^f - c_{jt}^f - \tau d_{jt}^f) \right] \quad (1)$$

Note, in this equation, Walmart is choosing to enter either a market g , f which already has a KMart, or a market with no firms. Entry is determined by profit net of costs in each location, discounted by a productivity adjusted discount factor.

The entry function for KMart is as follows:

$$P(\text{Entry})^K = \max_a \sum_{t=0}^{\infty} (\rho_t \beta)^t \left[\sum_{j \in \mathcal{B}_t^W} (\pi_{jt}^g - c_{jt}^g - \tau d_{jt}^g) + \sum_{j \in \mathcal{B}^0} (\pi_{jt}^f - c_{jt}^f - \tau d_{jt}^f) \right] \quad (2)$$

Same as above.

3 Simple Estimation

This section will present the results of the simple estimation techniques, probit under two specifications, and an IV regression. The probit specifications will differ by inclusion of competitor as a regressor. In the IV specification, I will use the log of the distance to Bentonville as an instrument for Walmart's entry in the KMart regression, and the number of small stores in the market as an instrument for KMart's entry in the Walmart regression.

3.1 Probit (No Competitor)

First, I will present the results of the probit regression without the competitor as a regressor.

$$P(\text{Entry}_j) = \Phi(\beta_0 + \beta_1 X_j) \quad (3)$$

where X_j is a vector of characteristics including log population, log retail sales per capita, percent urban, indicators for region, log distance to Bentonville (Walmart spec. only), and number of small stores

Walmart:

	Coefficient	Std. Error
Constant	-10.7057	1.037
Log Population	1.7870	0.094
Log Retail Sales (pc)	1.4514	0.119
% Urban	0.9521	0.197
Midwest	-0.1739	0.156
Log Distance to Bentonville	-1.1322	0.085
# Small Stores	-0.0701	0.019
South	0.6751	0.153

KMart:

	Coefficient	Std. Error
Constant	-18.8344	1.239
Log Population	1.4425	0.106
Log Retail Sales (pc)	1.4964	0.135
% Urban	1.0844	0.232
Midwest	0.4124	0.145
# Small Stores	0.0055	0.019
South	0.0546	0.142

From these results, we can see that Walmart faces less baseline entry friction than Kmart, as the constant is less negative. Population effects are also stronger for Walmart, as is being in the South. The distance to Bentonville is significantly negative, implying Walmart does not want to be too close to its headquarters, which is consistent with a belief of Walmart wanting to expand into new markets. Kmart seems to have a slight gain in the Midwest and a slight boost in entry with markets who are more urban.

3.2 Probit (Competitor)

Next, I will present the results of the probit regression with the competitor as a regressor.

Walmart:

	Coefficient	Std. Error
Constant	-12.9482	1.133
Log Population	1.9728	0.101
Log Retail Sales (pc)	1.6267	0.125
% Urban	1.1248	0.202
Midwest	-0.0554	0.157
Log Distance to Bentonville	-1.1002	0.086
# of Small Stores	-0.0677	0.020
South	0.7513	0.154
KMart	-0.6573	0.113

KMart:

	Coefficient	Std. Error
Constant	-21.5571	1.340
Log Population	1.7509	0.120
Log Retail Sales (pc)	1.7245	0.141
% Urban	1.2322	0.235
Midwest	0.5770	0.149
# Small Stores	-0.0016	0.020
South	0.3305	0.150
Walmart	-0.7102	0.111

Under this specification, the presence of a competitor has a significant negative effect on entry for both Walmart and KMart, but the effect is much stronger for KMart. This is consistent with the idea that Walmart is the dominant player in the market, and KMart is more likely to be deterred by Walmart's presence than vice versa.

3.3 IV Regression

Next, I will present the results of the IV regression. The results are as follows:

$$P(Entry_j) = \Phi(\beta_0 + \beta_1 X_j + \beta_2 Competitor_k) \quad (4)$$

where X_j is a vector of characteristics including log population, log retail sales per capita, percent urban, indicators for region, log distance to Bentonville (Walmart spec. only), and number of small stores. The competitor variable is instrumented as described above.

$$P(Entry_i) = \Phi(\beta_0 + \beta_1 X_i + \beta_2 \hat{Competitor}_k) \quad (5)$$

Walmart:

	Coefficient	Std. Error
Constant	-10.6671	nan
Log Population	1.7870	0.094
Log Retail Sales (pc)	1.4514	0.119
% Urban	0.9521	0.197
Midwest	-0.1739	0.156
Log Distance to Bentonville	-0.0459	nan
South	0.6751	0.153
$\hat{K}Mart$	-0.6425	nan

KMart:

	Coefficient	Std. Error
Constant	-17.4435	1.260
Log Population	1.4670	0.108
Log Retail Sales (pc)	1.4752	0.137
% Urban	1.2727	0.239
Midwest	1.1634	0.173
South	0.7797	0.169
# Small Stores	0.0222	0.020
<i>Walmart</i>	-4.5786	0.581

The IV results show that Walmart's predicted entry has a massively negative effect on KMart's entry probability, implying that a) Walmart's entry is a strong deterrent to KMart's entry, and b) the OLS results were biasing the results downwards. For Walmart, KMart's predicted entry has a negative effect, but there is likely a misspecification in instrument as standard errors are not available. To be diagnosed. However, the general magnitude is consistent with the OLS results, implying that KMart is less of a deterrent to Walmart.

3.4 Comparing Across Specifications

In the simple case, it would seem we are biasing the results upwards by not including competition effects, as the presence of a competitor should act as a deterrent. When we include competition effects, we see that the constant term becomes more negative, which is more consistent with the idea that there are significant barriers to entry, and the presence of a competitor is a significant deterrent to entry. The IV results show that the presence of a competitor is an even stronger deterrent than the OLS results suggest, which implies that there is likely some endogeneity in the OLS results, as firms are more likely to enter into markets where there are no competitors, and less likely to enter into markets where there are competitors. This effect is especially pronounced for KMart, who sees Walmart's potential entry as a strong deterrent.

4 Ordered Probit

4.1 Specification 1

In the first specification of the ordered probit, we group Walmart and Kmart together, and model the number of firms in the market as a function of the same characteristics as above. The results are as follows:

	Coefficient	Std. error
Log Population	1.9054	0.080
Log Retail Sales (pc)	1.6348	0.101
% Urban	1.2092	0.168
Midwest	0.4985	0.132
Log Distance to Bentonville	-0.3384	0.056
# Small Stores	-0.0451	0.015
South	0.8400	0.131
Effect of $0 \rightarrow 1$	17.8011	0.937
Effect of $1 \rightarrow 2$	0.8412	0.032

Here, we see that being the first entrant has a significant positive effect on entry, but the second entrant is a fraction of the effect, though still significant. This result lends itself to the idea of first mover advantages, as the first entrant is able to capture a large portion of the market. We still see a negative effect of distance to Bentonville, but the effect is smaller than in the simple probit, which is consistent with the idea that distance to Bentonville is more of a deterrent for Walmart than KMart. We also see that as there are more small stores, there is a deterrance to entry, though this effect is small.

4.2 Specification 2

Here, we group all firms together, modeling number of firms in the market as a function of the same characteristics as above. The results are as follows:

	Coefficient	Std Error
Log Population	1.7589	0.073
Log Retail Sales (pc)	1.2985	0.085
% Urban	1.1849	0.152
Midwest	0.5019	0.124
Log Distance to Bentonville	-0.2866	0.053
# Small Stores	2.7925	0.071
South	0.7867	0.122
0 → 1 Firm	14.0616	0.801
1 → 2 Firms	1.2129	0.047
2 → 3 Firms	1.0088	0.042
3 → 4 Firms	1.0971	0.041
4 → 5 Firms	1.0198	0.045
5 → 6 Firms	1.0504	0.049
6 → 7 Firms	1.0265	0.053
7 → 8 Firms	0.9840	0.064
8 → 9 Firms	0.8639	0.080
9 → 10 Firms	1.2793	0.068
10 → 11 Firms	0.8734	0.119
11 → 12 Firms	1.0653	0.109
12 → 13 Firms	0.9233	0.121

The first mover advantage is still present, but to a lesser degree. Second mover advantage has also increased. Interestingly, after being the first mover, the number of firms in the market has a relatively stable effect on entry, implying that only first mover advantage is important, and after that, the number of firms in the market does not have a major effect on entry. Distance to Bentonville is still negative, but less so than in the above. Small stores are now positive, which makes sense given small stores are more likely to enter into markets where other small firms are present.

5 Seim Method

In this section, I will present the results of the Seim method. The results are as follows:

Variable	Walmart Coefficient	KMart Coefficient
Log Population	6.68	1.25
Log Retail Sales (pc)	0.11	-1.03
Percent Urban	8.95	3.10
Midwest	-1.63	0.10
Log Distance to Bentonville	-3.28	0.17
Number of Small Stores	0.06	0.10
South	-1.55	-0.82
Predicted KMart Entry	1.15	—
Preidcted Walmart Entry	—	-14.38

These results show that Walmart still has a strong effect on KMart’s entry probability, similar in magnitude to the constant from the simple probit regression. Where I think I have made an error, is that Walmart sees Kmart’s potential entry as a positive effect, though very small. This is counterintuitive, but if I were to rationalize it, I would say that Walmart sees Kmart’s potential entry as a sign of a healthy market, and because Walmart is the dominant firm, it is not deterred, but rather motivated to enter. However, I will have to go back to my code and check for bugs, as this result is not what I expected. Besides the Δ_i terms, the results are generally consistent with the other specifications, with population and percent urban being strong predictors of entry, and distance to Bentonville being a strong detterent for Walmart.

6 Conclusion

In conclusion, this problem set was concerned with evaluating models of entry and competition. Going from the simplest possible probit model of entry to a (simplified) method which accounts for strategic planning of entry, we see that the effect of a competitor on the entry decision is significant, and likely much stronger than can be fully captured in a simple probit. Comparing across models, one thing is constant: a competitor entering the market is a deterrent to entry. What differs is the magnitude of the effect. In the probits, this is a very small effect, with the IV having the strongest at -4.58 for predicted Walmart entry on KMart. When we move to the ordered probit, we see that the first mover is greatly incentivized to enter, but after that, the gain from entry is much smaller, consistent with the idea that being first to market provides great value. Finally, the Seim method shows that Walmart’s predicted entry is hugely influential on KMart’s entry decision, but KMart’s predicted entry may actually slightly encourage Walmart to enter the market, though this may be a bug.

In all, I think the ordered probit and Seim methods are more consistent with the idea of strategic entry, and would be used well in tandem to model the effect of competition on entry. The simple probit methods are useful for baseline estimation, but the true effect of entry competition is likely

stronger than can be captured.

A Code

A.1 Main Code

```
import numpy as np
import scipy as sp
import statsmodels.api as sm
import pandas as pd
from prettytable import PrettyTable
from statsmodels.miscmodels.ordinal_model import OrderedModel

"""
Problem_Set_2:

Author:_Tate_Mason
Due:_04-23-2026
Class:_IO_2

_*_Helper_files:
    *_PS2c.py:_Contains_the_implementation_of_the_Seim_approach_for_estimating_entry_
        decisions_in_a_duopoly_setting.

_*_Libraries_used:
    *_numpy
    *_scipy
    *_statsmodels
    *_pandas
    *_prettytable
"""

# Read Data

xmat_cols = [
    'county_id', 'log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
        log_dist_benton', 'south',
    'kmart', 'walmart', 'n_small_stores', 'dw_kmart_out', 'dw_walmart_out', 'col13', '
        col14', 'col15'
]

XMat = pd.read_csv('../Data/XMat.out', sep=r'\s+', header=None, names=xmat_cols) #
    read in the data, specifying column names and separator
print(XMat.head()) # checking first few rows of data

# Helper function to save LaTeX tables
import os
os.makedirs('../Tables', exist_ok=True)
def save_tex_table(latex_str, filename):
```

```

    with open(f'../Tables/{filename}.tex', 'w') as f:
        f.write(latex_str)

#### Problem 2a ===#

"""
Here we will use a simple probit under different specifications to estimate the entry
decision of a firm

- Data:
-----XMat.out: County level data on various store level controls.
"""

from PS2a import *

#### Part A ===#

print("===_Part_A_===")

lhs_wm = XMat['walmart'] # dependent variable for Walmart entry
rhs_wm = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    log_dist_benton', 'n_small_stores', 'south']] # independent variables for Walmart
    entry

wm_probit = sm.Probit(lhs_wm, sm.add_constant(rhs_wm)).fit() # fit the probit model
    for Walmart entry
save_tex_table(wm_probit.summary().as_latex(), 'wm_probit_spec1') # save the summary
    of the model as a LaTeX table

lhs_km = XMat['kmart'] # dependent variable for Kmart entry
rhs_km = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'south']] # independent variables for Kmart entry

km_probit = sm.Probit(lhs_km, sm.add_constant(rhs_km)).fit() # fit the probit model
    for Kmart entry
save_tex_table(km_probit.summary().as_latex(), 'km_probit_spec1') # save the summary
    of the model as a LaTeX table

#### Part B ===#

print("===_Part_B_===")

lhs_wm_comp = XMat['walmart'] # dependent variable WM entry
rhs_wm_comp = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    log_dist_benton', 'n_small_stores', 'south', 'kmart']] # independent variables for
    WM entry, including Kmart entry as a regressor

wm_comp_probit = sm.Probit(lhs_wm_comp, sm.add_constant(rhs_wm_comp)).fit() # fit

```

```

    model
save_tex_table(wm_comp_probit.summary().as_latex(), 'wm_probit_spec2') # save summary
    as LaTeX table

lhs_km_comp = XMat['kmart'] # dependent variable KM entry
rhs_km_comp = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'south', 'walmart']] # independent variables for KM entry,
    including WM entry as a regressor

km_comp_probit = sm.Probit(lhs_km_comp, sm.add_constant(rhs_km_comp)).fit() # fit
    model
save_tex_table(km_comp_probit.summary().as_latex(), 'km_probit_spec2') # save summary
    as LaTeX table

# The biggest weakness of using the number of competitor stores as a regressor is the
    almost assured endogeneity. The presence of a competitor store is likely correlated
    with unobserved factors that also affect the entry decision, such as local market
    conditions, consumer preferences, or the strategic behavior of firms.
# For example, if a county has favorable market conditions for retail entry (e.g.,
    high population density, strong consumer demand), both Walmart and Kmart may be
    more likely to enter, leading to a positive correlation between the presence of one
    firm and the other. This endogeneity can bias the estimated coefficients and lead
    to incorrect inferences about the competitive effects on entry decisions.

#=== Part C ===#
print("===_Part_C_===")

lhs_km_iv = XMat['kmart'] # dependent variable for Kmart entry
rhs_km_iv = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'south', 'walmart']] # independent variables for Kmart entry,
    including Walmart entry as a regressor
instrument_km_iv = 'log_dist_benton' # IV for Walmart entry in Kmart equation

# First stage regression
first_stage_km_iv = sm.OLS(XMat['walmart'], sm.add_constant(XMat[instrument_km_iv])).
    fit() # fit the first stage regression of Walmart entry on the instrument (distance
    from Benton Co.)
XMat['predicted_walmart'] = first_stage_km_iv.predict(sm.add_constant(XMat[
    instrument_km_iv])) # generate predicted values of Walmart entry from the first
    stage regression

# Second stage regression

rhs_km_iv2 = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'south',
    'n_small_stores', 'predicted_walmart']] # use the predicted values of Walmart entry
    in the second stage regression for Kmart entry
model_km_iv = sm.Probit(lhs_km_iv, sm.add_constant(rhs_km_iv2)) # fit the second stage
    regression using the predicted values of Walmart entry as a regressor in the Kmart

```

```

    entry equation
results_km_iv = model_km_iv.fit() # fit the model and get results
print(results_km_iv.summary()) # print the summary of the second stage regression for
    Kmart entry with IV
save_tex_table(results_km_iv.summary().as_latex(), 'km_iv_probit') # save the summary
    of the second stage regression for Kmart entry with IV as a LaTeX table

lhs_wm_iv = XMat['walmart'] # dependent variable for Walmart entry
rhs_wm_iv = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'log_dist_benton', 'south', 'kmart']] # independent variables for
    Walmart entry, including Kmart entry as a regressor
instrument_wm_iv = 'n_small_stores' # IV for Kmart entry in Walmart equation

# First stage regression
first_stage_wm_iv = sm.OLS(XMat['kmart'], sm.add_constant(XMat[instrument_wm_iv])).fit
    () # fit the first stage regression of Kmart entry on the instrument (number of
    small stores)
XMat['predicted_kmart'] = first_stage_wm_iv.predict(sm.add_constant(XMat[
    instrument_wm_iv])) # generate predicted values of Kmart entry from the first stage
    regression

# Second stage regression
rhs_wm_iv2 = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', '
    n_small_stores', 'log_dist_benton', 'south', 'predicted_kmart']] # use the
    predicted values of Kmart entry in the second stage regression for Walmart entry
model_wm_iv = sm.Probit(lhs_wm_iv, sm.add_constant(rhs_wm_iv2)) # fit the second stage
    regression using the predicted values of Kmart entry as a regressor in the Walmart
    entry equation
results_wm_iv = model_wm_iv.fit() # fit the model and get results for the second stage
    regression for Walmart entry with IV
print(results_wm_iv.summary()) # print the summary of the second stage regression for
    Walmart entry with IV
save_tex_table(results_wm_iv.summary().as_latex(), 'wm_iv_probit') # save the summary
    of the second stage regression for Walmart entry with IV as a LaTeX table

# Kmart IV results suggest that coeff on pred. WM is significant and negative. This
    implies that as distance from benton co. increases, the probability of Walmart
    entry increases, which in turn reduces the probability of Kmart entry.
# This makes sense, as WM would be more likely to enter as they stray from their home
    county. The exclusion restriction seems plausible, as distance from Benton Co.
    should not directly affect Kmart entry decisions, except through its effect on
    Walmart's entry.
# Relevance condition is satisfied as distance from Benton Co. is likely correlated
    with Walmart's entry decision, given that Benton Co. is Walmart's home county and a
    central location for their operations.
#
# WM IV coef on pred. KM is significant and negative, suggesting that as the number of
    small stores increases, the probability of Kmart entry increases, which in turn

```

```

    reduces the probability of Walmart entry. This also makes sense, as a higher number
    of small stores may indicate a more competitive retail environment, making it less
    attractive for Walmart to enter.
# The exclusion restriction seems implausible, as the number of small stores could
    directly affect Walmart's entry decision, independent of Kmart's entry. For example
    , a high number of small stores could indicate a saturated market, which would
    deter Walmart from entering regardless of Kmart's presence.
# The relevance condition is satisfied as the number of small stores is likely
    correlated with Kmart's entry decision, as Kmart may be more likely to enter
    markets with fewer small stores where they can capture more market share.

#### Bresnahan-Reiss Approach ####

# Ordered Probit

# Spec 1: LHS = number of large players, RHS = market controls

controls = ['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'log_dist_benton',
            ', 'n_small_stores', 'south'] # list of control variables for the ordered probit
            regression

XMat['n_large'] = XMat['kmart'].astype(int) + XMat['walmart'].astype(int) # create a
            new variable for the number of large players (0, 1, or 2) by summing the binary
            indicators for Kmart and Walmart entry

# Fit the ordered probit model using the number of large players as the dependent
            variable and the market controls as independent variables
res_spec1 = OrderedModel(
            XMat['n_large'], XMat[controls], distr='probit'
).fit(method='BFGS')

print(res_spec1.summary())
save_tex_table(res_spec1.summary().as_latex(), 'ordered_probit_spec1') # save the
            summary of the ordered probit model for Spec 1 as a LaTeX table

# Spec 2: LHS = number of players (large + small), RHS = market controls

XMat['n_total'] = XMat['n_large'] + XMat['n_small_stores'] # create a new variable for
            the total number of players (large + small) by summing the number of large players
            and the number of small stores

# Fit the ordered probit model using the total number of players as the dependent
            variable and the market controls as independent variables
res_spec2 = OrderedModel(
            XMat['n_total'], XMat[controls], distr='probit'
).fit(method='BFGS')

print(res_spec2.summary())

```

```

save_tex_table(res_spec2.summary().as_latex(), 'ordered_probit_spec2') # save the
    summary of the ordered probit model for Spec 2 as a LaTeX table

=== Seim Approach ===

from PS2c import * # import the functions for the Seim approach from the helper file
    PS2c.py

# Prepare data for Seim approach
XW = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'log_dist_benton',
    , 'n_small_stores', 'south']].values
XK = XMat[['log_pop', 'log_retail_sales_pc', 'pct_urban', 'midwest', 'n_small_stores',
    'log_dist_benton', 'south']].values

yW = XMat['walmart'].values
yK = XMat['kmart'].values

K = XW.shape[1]

theta0 = np.zeros(2*K + 2) # initial parameter guess

res_seim = minimize(log_lik, theta0, args=(XW, XK, yW, yK), method='BFGS') # optimize
    log likelihood using BFGS

# Extract parameter estimates
theta_hat = res_seim.x
betaW_hat = theta_hat[:K]
deltaW_hat = theta_hat[K]
betaK_hat = theta_hat[K+1:2*K+1]
deltaK_hat = theta_hat[-1]

# Display results in a table
res_table = PrettyTable()
res_table.field_names = ["Parameter", "Estimate"]
for i in range(K):
    res_table.add_row([f"betaW_{i}", betaW_hat[i]])
res_table.add_row(["deltaW", deltaW_hat])
for i in range(K):
    res_table.add_row([f"betaK_{i}", betaK_hat[i]])
res_table.add_row(["deltaK", deltaK_hat])
print(res_table)
param_names = [f'betaW_{i}' for i in range(K)] + ['deltaW'] + \
    [f'betaK_{i}' for i in range(K)] + ['deltaK']

seim_df = pd.DataFrame({
    'Parameter': param_names,

```



```

        'Estimate': theta_hat
    })

# Export the Seim results to a LaTeX table
save_tex_table(
    seim_df.to_latex(index=False, float_format='%.4f', caption='Seim_NFP_Estimates',
        label='tab:seim'),
    'seim_results'
)

```

A.2 Seim Code

```

import numpy as np
import scipy as sp
from scipy.special import expit
from scipy.optimize import minimize

def solve_eq(xW, xK, theta, eps=1e-6, max_iter=1000):

    betaW = theta[:xW.shape[1]] # Walmart coefficients
    deltaW = theta[xW.shape[0]] # Walmart's influence on KMart

    betaK = theta[xW.shape[0]+1 : xW.shape[0] + 1 + xK.shape[0]] # KMart coefficients
    deltaK = theta[-1] # KMart's influence on Walmart

    PW, PK = 0.5, 0.5 # initial guess

    for _ in range(max_iter):
        PW_new = expit(xW@betaW + deltaW*PK) # update PW based on current PK
        PK_new = expit(xK@betaK + deltaK*PW) # update PK based on current PW

        if max(abs(np.log(PW_new / PW)), abs(np.log(PK_new / PK))) <= eps:
            return PW_new, PK_new # convergence achieved
        PW, PK = PW_new, PK_new # update for next iteration

    return PW, PK # return last estimates if max iterations reached without
    convergence

def log_lik(theta, XW, XK, yW, yK):
    M = XW.shape[0] # number of markets
    ll = 0.0 # log-likelihood initialization

    for m in range(M):
        PW, PK = solve_eq(XW[m], XK[m], theta) # call above function to get PW and PK
        for market m

        PW = np.clip(PW, 1e-10, 1 - 1e-10) # to avoid log(0)
        PK = np.clip(PK, 1e-10, 1 - 1e-10) # to avoid log(0)

```

```
ll += yW[m]*np.log(PW) + (1 - yW[m])*np.log(1 - PW) # log-likelihood
      contribution from Walmart
ll += yK[m]*np.log(PK) + (1 - yK[m])*np.log(1 - PK) # log-likelihood
      contribution from KMart
return -ll # return negative log-likelihood for minimization
```